

Aastha Mittal

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EDUCATION

Carnegie Mellon University

B.S. Computer Science, Concentrations: Systems & Machine Learning/AI

Pittsburgh, PA

Expected: May 2028

Relevant Coursework: Machine Learning, Parallel & Sequential Data Structures & Algorithms, Computer Systems, Probability & Statistics

WORK EXPERIENCE

Software Engineer Intern

Jan 2026 – Present

Time2Cash (Legal Tech Startup, \$5M Seed Round)

Remote

- Designed and implemented scalable microservices in Python handling 50K+ daily requests at 99.2% uptime; applied data structures and algorithms to build efficient processing pipelines using PostgreSQL, Redis, and Kafka
- Built LLM-powered document analysis system (OpenAI API) with Pandas/NumPy post-processing and statistical anomaly detection, reducing manual review time by 65% and surfacing \$2M+ in revenue leakage

Teaching Assistant — 15-121: Introduction to Data Structures

Aug 2025 – Present

Carnegie Mellon University

Pittsburgh, PA

- Teaching data structures and algorithms in Java to 25+ students through weekly lab sections; provide one-on-one debugging support to reinforce fundamental computer science concepts

Software Engineer Intern

May 2025 – Aug 2025

Nationwide Insurance

Columbus, OH

- Developed large-scale full-stack application in JavaScript/TypeScript and Java serving 1M+ users; wrote clean, well-tested code with 90%+ coverage and participated in code reviews to uphold engineering standards
- Designed and implemented CI/CD infrastructure with Jenkins reducing deployment time from 2 hours to 15 minutes across 5+ environments; collaborated with cross-functional teams to ship user-facing features

Software Engineer Intern

May 2024 – Aug 2024

Nationwide Insurance

Scottsdale, AZ

- Built backend testing framework with 200+ unit tests in Java achieving 90% code coverage; integrated observability tooling into 5+ internal systems, reducing defect reproduction time by 75%

Software Engineer Intern

Jun 2023 – Jul 2023

Arway Inc.

Scottsdale, AZ

- Implemented backend API components for high-traffic platform (500K+ users) with OAuth 2.0/JWT auth; contributed to distributed gateway architecture supporting 10K+ req/sec with rate limiting and load balancing

PROJECTS

LLM Knowledge Retrieval & Q&A System | Python, LangChain, OpenAI API, FAISS, PostgreSQL

2025

- Designed and built a retrieval-augmented generation system using vector search (FAISS) and OpenAI embeddings; engineered efficient indexing and retrieval algorithms to enable accurate Q&A over large document corpora
- Instrumented precision and MRR metrics for system observability; implemented human-in-the-loop review for low-confidence outputs — end-to-end ownership from design through deployment

MedEase: Healthcare Made Easy | React, TypeScript, Python, GCP, Firebase, SQL

Aug 2023 – May 2024

- Architected and shipped end-to-end multilingual web application: React/TypeScript frontend, Python backend, Firebase real-time database, and cloud ML infrastructure — serving 100+ languages with 92% accuracy
- Designed scalable data pipelines and intent classification system with anomaly detection; applied software engineering best practices including code reviews, testing, and CI/CD throughout development lifecycle

FitForm Analyzer | Python, PyTorch, NumPy, REST API, Computer Vision

Nov – Dec 2024

- Trained and optimized neural network achieving 88% accuracy for real-time posture classification across 5+ exercise types; applied model compression techniques to reduce inference latency by 40%
- Processed 100K+ joint position data points using cosine similarity and computer vision algorithms; exposed predictions via REST API with rigorous statistical validation of output robustness

TECHNICAL SKILLS

Languages: Python, Java, JavaScript/TypeScript, C/C++, SQL, HTML/CSS

Data Structures & Algorithms: Proficient — applied in production systems, competitive problem solving, and coursework

Systems & Infrastructure: REST APIs, microservices, distributed systems, Docker, Kubernetes, CI/CD, Linux

Databases & Data: PostgreSQL, MySQL, Redis, Kafka, FAISS — schema design, query optimization, pipeline engineering

ML & AI: PyTorch, OpenAI API, LangChain, RAG/Vector Search, NumPy, Pandas, Scikit-learn, TensorFlow

LEADERSHIP & ACTIVITIES

CMU ScottyLabs (CMU Maps) — Software Developer (Aug 2024–Present) | **Jane Street Discovery Day** | NCWIT

National Winner (Apr 2024)

Professional Speaker — Women in Tech & EdTech (50+ events) | CMU Quant Club | CMU Hindu Yuva — Community Outreach Head